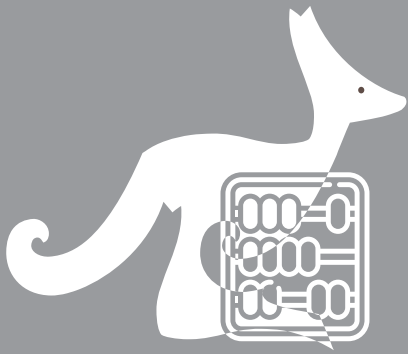




MATH KANGAROO CHINA

考试说明 / Instruction

Time allowed: 75 minutes

考试时长: 75 分钟

Language: English & Chinese

试卷语言: 英文和中文

This is a 30-question multiple-choice competition.

考试包含 30 道单项选择题。

Problems 1 to 10 are worth 3 points each.

1-10 题每题 3 分

Problems 11 to 20 are worth 4 points each.

11-20 题每题 4 分

Problems 21 to 30 are worth 5 points each.

21-30 题每题 5 分

1 point will be deducted for an incorrect answer and
no point for a blank answer.

答错扣 1 分, 不答不扣分。

The full score is 150 (Starting point is 30)

总分为 150 分 (起始分数为 30 分)

Do not open this booklet until you are told to do so.

收到监考老师指令后, 方可打开试卷册。

Mark your answers clearly on the answer sheet using a **2B pencil**.

务必使用 **2B 铅笔**在答题卡上正确填涂自己的答案。

Calculator cannot be used during the competition.

考试期间禁止使用计算器。

DO NOT discuss problems and solutions in any online or public
forum until April 30.

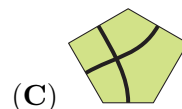
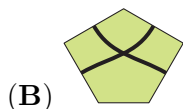
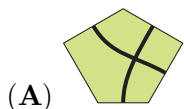
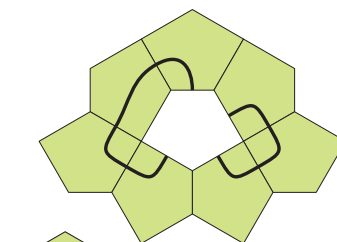
4 月 30 日前请勿以任何形式公开讨论试题和答案。

**Grade 11
&
Grade 12**

3 points

1. A pattern is made of equal pentagons. Which of the tiles below, when placed in the central hole, will form a self-intersecting loop?

右侧图形由相同的五边形瓷砖组成。请问将下列哪块瓷砖放入图形的中心处可以形成一条自相交叉的环线？



2. Which of these integers is two less than a multiple of ten, two more than a square, and two times a prime?

请问下列哪一个整数比 10 的倍数少 2，比一个平方数多 2，并且是一个质数的两倍？

(A) 78

(B) 58

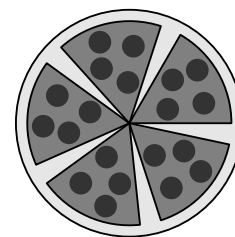
(C) 38

(D) 18

(E) 6

3. A young kangaroo cut a pizza into six equal slices. After eating one slice, he arranged the remaining slices with equal gaps between slices. What size is the angle of each gap?

一只小袋鼠将披萨分为 6 等份。小袋鼠吃了一份之后，将剩余的披萨摆放整齐，使得每份披萨之间的夹角相等。请问相邻披萨之间的夹角为多少度？



(A) 5°

(B) 8°

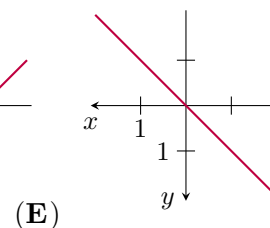
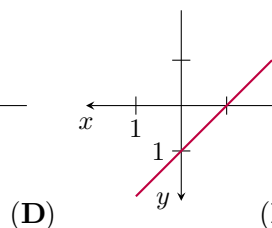
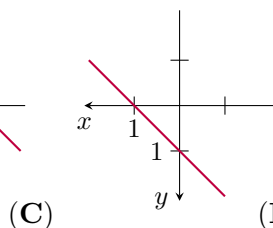
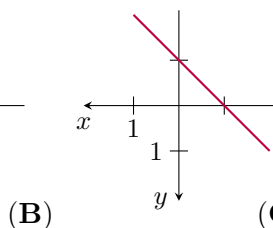
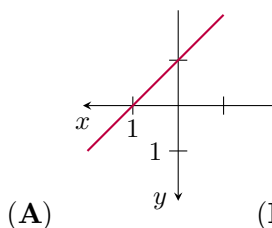
(C) 9°

(D) 10°

(E) 12°

4. Juuso has an unusual habit of drawing the xy -plane with the positive coordinate axes pointing left and down. What would the graph of the equation $y = x + 1$ look like in a coordinate system drawn by Juuso?

Juuso 画 xy 平面直角坐标系时，习惯将向左和向下的方向定为坐标轴的正方向。请问下列哪一项是方程 $y = x + 1$ 在 Juuso 画的 xy 平面直角坐标系上的图像？



5. Kaito has manipulated a die. The probabilities of rolling a 2, 3, 4 or 5 are still $\frac{1}{6}$ each, but the probability of rolling a 6 is twice the probability of rolling a 1. What is the probability of rolling a 6?

Kaito 操控一枚骰子, 使得点数 2、3、4、5 中各个点数朝上的概率始终为 $\frac{1}{6}$, 但是点数 6 朝上的概率是点数 1 朝上的概率的两倍。请问点数 6 朝上的概率是多少?

- (A) $\frac{1}{4}$ (B) $\frac{1}{6}$ (C) $\frac{7}{36}$ (D) $\frac{2}{9}$ (E) $\frac{5}{18}$

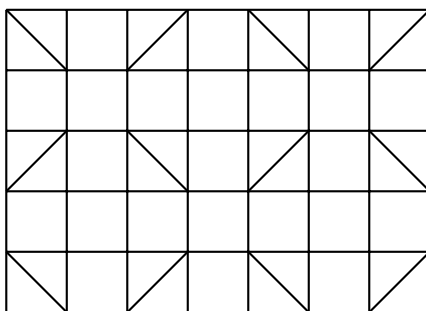
6. Which of the expressions below has the same value as $16^{15} + 16^{15} + 16^{15} + 16^{15}$?

请问下列哪个表达式与 $16^{15} + 16^{15} + 16^{15} + 16^{15}$ 的值相等?

- (A) 16^{19} (B) 4^{31} (C) 4^{60} (D) 16^{60} (E) 4^{122}

7. Beaver wishes to color the squares and triangles of the following figure so that no two neighbouring figures, even those sharing a single vertex, are the same color.

Beaver 想为下图中的正方形和三角形上色, 使得任意两个相邻图形不同色, 包括共用一个顶点的两个图形也不同色。



What is the least number of colors needed?

请问 Beaver 至少需要几种颜色?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

8. There are 6 glasses on a table with their open ends up. In any one move, we turn over exactly 4 of them. What is the least number of moves required to have all glasses upside down?

桌上有 6 个开口朝上的玻璃杯。每一轮只翻转 4 个玻璃杯, 请问至少需要几轮才能使得所有玻璃杯开口朝下?

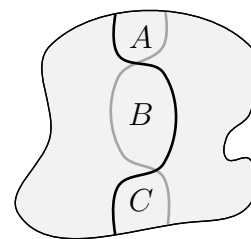
- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

9. A student started with the number 1 and multiplied it by either 6 or 10. He then multiplied the result by either 6 or 10, and continued this procedure many times. Which of the following cannot be one of the numbers he obtained?

一名学生从数字 1 开始, 将其乘 6 或 10, 然后再将得到的结果乘 6 或 10, 以此类推, 多次重复该过程。请问这名学生无法得到下列哪一个数字?

- (A) $2^{100}3^{20}5^{80}$ (B) $2^{90}3^{20}5^{80}$ (C) $2^{90}3^{20}5^{70}$ (D) $2^{110}3^{80}5^{30}$ (E) $2^{50}5^{50}$

10. A black trail and a grey trail cross a park, as shown. Each trail divides the park into two regions of equal area. Which of the following must be true about the areas A , B and C ?



如图所示，一条黑色小道和一条灰色小道横穿公园。每一条小道都将公园划分为两个面积相同的区域。请问关于面积 A 、 B 、 C 的说法中，下列哪项是正确的？

- (A) $A = C$ (B) $B = A + C$ (C) $B = \frac{1}{2}(A + C)$
 (D) $B = \frac{2}{3}(A + C)$ (E) $B = \frac{3}{5}(A + C)$

4 points

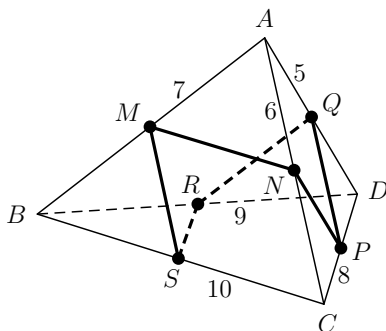
11. Exactly one of these statements about a certain positive integer n is true. Which statement is true?

下列关于某个正整数 n 的表述中恰好只有一项正确，请问是哪一项？

- (A) n is divisible by 3 n 可以被 3 整除 (B) n is divisible by 6 n 可以被 6 整除
 (C) n is odd n 是奇数 (D) $n = 2$
 (E) n is prime n 是质数

12. A triangular pyramid $ABCD$ has sides of length 5, 6, 7, 8, 9 and 10. The points M , N , P , Q , R and S are the midpoints of the edges of the pyramid, as shown.

三棱锥 $ABCD$ 的各边长为 5、6、7、8、9、10。点 M 、 N 、 P 、 Q 、 R 、 S 是三棱锥各边上的中点，如图所示。

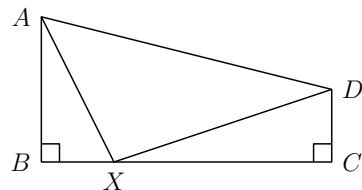


What is the perimeter of the closed hexagonal line $MNPQRSM$?

请问闭合的六边形 $MNPQRSM$ 的周长是多少？

- (A) 19 (B) 20 (C) 21 (D) 22 (E) 23

13. A quadrilateral $ABCD$ has two right angles at B and C , where $AB = 4$, $BC = 8$ and $CD = 2$. Point X lies on BC . What is the minimum value of $AX + DX$?



四边形 $ABCD$ 有两个直角位于 B 和 C 处，且 $AB = 4$, $BC = 8$, $CD = 2$ 。点 X 位于 BC 边上。请问 $AX + DX$ 的最小值是多少？

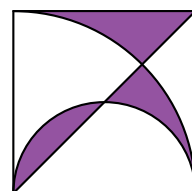
- (A) $9\sqrt{2}$ (B) 12
(C) 13 (D) 10
(E) None of the previous 以上都不正确

14. John has a number of all black or all white unit cubes and wants to build a $3 \times 3 \times 3$ cube using 27 of them. He wants the surface to be exactly half black and half white. What is the smallest number of black cubes he can use?

John 有很多全黑和全白的单位立方体。他想用其中 27 个单位立方体构建一个 $3 \times 3 \times 3$ 的大立方体。他希望这个大立方体的表面上，恰好一半为黑色，一半为白色，请问 John 最少需要用多少个全黑单位立方体？

- (A) 14 (B) 13 (C) 12 (D) 11
(E) None of the previous 以上都不正确

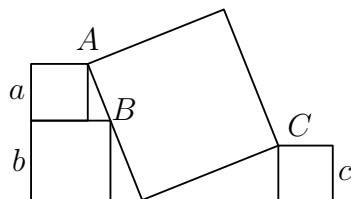
15. A diagonal, a semicircle and a quadrant are drawn in a square of side 6 cm. What is the area, in cm^2 , of the shaded part?



在一个边长为 6 cm 的正方形中画一条对角线、一个半圆和一个四分之一圆。请问阴影部分的面积为多少 cm^2 ？

- (A) 9 (B) 3π (C) $6\pi - 9$ (D) $10\pi/3$ (E) 12

16. The figure shows four squares. The smaller ones have side lengths a , b and c . The vertices A and C of two of the smaller squares coincide with two diagonally opposite vertices of the large square. The vertex B of the third small square is on the side of the large one. Which of the following expressions represents the side length of the largest square?



右图中有 4 个正方形。三个小正方形的边长分别为 a 、 b 、 c 。其中两个小正方形的顶点 A 和 C 恰好是最大正方形的两个对角顶点，第三个小正方形的顶点 B 落在最大正方形的边上。请问下列哪一项表示最大正方形的边长？

- (A) $\frac{1}{2}(a + b + c)$ (B) $\sqrt{a^2 + b^2 + c^2}$ (C) $\sqrt{(a + b)^2 + c^2}$
(D) $\sqrt{(b - a)^2 + c^2}$ (E) $\sqrt{a^2 + ab + b^2 + c^2}$

17. We have two positive numbers p and q , with $p < q$. Which of these expressions is the largest?

p 和 q 为两个正数, 且 $p < q$ 。请问下列哪个表达式的值最大?

- (A) $\frac{p+3q}{4}$ (B) $\frac{p+2q}{3}$ (C) $\frac{p+q}{2}$ (D) $\frac{2p+q}{3}$ (E) $\frac{3p+q}{4}$

18. How many three-digit numbers are there that contain at least one of the digits 1, 2 or 3?

请问有多少个 3 位数, 其数位上至少包含 1、2、3 这三个数字中的一个?

- (A) 27 (B) 147 (C) 441 (D) 557 (E) 606

19. I write down a 4-digit non-zero number $N = \overline{pqrs}$. When I place a decimal point between the q and the r , I find that the resulting number $\overline{pq.r\overline{s}}$ is the average of the two-digit numbers $\overline{pq}$ and $\overline{rs}$.

What is the sum of the digits of N ?

写下一个非零的四位数 $N = \overline{pqrs}$ 。在 q 和 r 之间加入一个小数点, 得到的数字 $\overline{pq.r\overline{s}}$ 是两位数 $\overline{pq}$ 和 $\overline{rs}$ 的平均值。请问数字 N 的各位数字之和为多少?

- (A) 14 (B) 18 (C) 21 (D) 25 (E) 27

20. Two candles of equal length start burning at the same time. One of the candles will burn down in 4 hours, the other in 5 hours, each at their own constant rate. How many hours will they have to burn before one candle is 3 times the length of the other?

两支等长的蜡烛同时开始燃烧。每支蜡烛的燃烧速度都是恒定的, 其中一支蜡烛 4 小时后燃尽, 另外一支 5 小时后燃尽。请问两支蜡烛必须燃烧多少小时之后, 其中一支蜡烛的长度是另一支蜡烛长度的 3 倍?

- (A) $\frac{40}{11}$ (B) $\frac{45}{12}$ (C) $\frac{63}{20}$ (D) 3 (E) $\frac{47}{14}$

5 points

21. Andre has six cards with one number written on each side of each card. The pairs of numbers on the cards are (5, 12), (3, 11), (0, 16), (7, 8), (4, 14) and (9, 10). The cards can be placed in any order in the blank spaces of the figure.

Andre 有 6 张卡片, 每张卡片的正反两面上各写有一个数字。6 张卡片上的数对分别为 (5, 12)、(3, 11)、(0, 16)、(7, 8)、(4, 14)、(9, 10)。将这 6 张卡片以任意顺序放入下方空格中。

$$\square + \square + \square - \square - \square - \square = ?$$

What is the smallest result he can get?

请问 Andre 可以得到的最小值是多少?

- (A) -23 (B) -24 (C) -25 (D) -26 (E) -27

22. Kangaroo solves the equation $ax^2 + bx + c = 0$, and Beaver solves the equation $bx^2 + ax + c = 0$, where a, b, c are pairwise distinct non-zero integers. It turns out that the equations share a solution. Which of the following must be true?

Kangaroo 解方程 $ax^2 + bx + c = 0$, Beaver 解方程 $bx^2 + ax + c = 0$, 其中 a, b, c 是互不相同的非零整数, 且两个方程有一个相同的解。请问下列哪项的说法一定是正确的?

(A) The common solution must be 0.

两个方程的公共解一定是 0。

(B) The quadratic equation $ax^2 + bx + c = 0$ has exactly one real solution.

二次方程 $ax^2 + bx + c = 0$ 只有一个实数解。

(C) $a > 0$

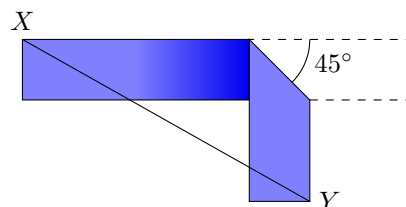
(D) $b < 0$

(E) $a + b + c = 0$

23. I have a strip of paper that is 12 cm long and 2 cm wide. I make a crease across it at 45° and then fold it, so that the two parts of the strip are aligned in a right angle, as shown.

What is the smallest possible length, in cm, of XY ?

一张纸条, 长 12 cm, 宽 2 cm。在纸条的 45° 处做一条折痕, 然后沿着折痕进行折叠, 此时纸条的两个部分呈直角, 如图所示。请问 XY 的长度最短可能为多少 cm?



(A) $6\sqrt{2}$

(B) $7\sqrt{2}$

(C) 10

(D) 8

(E) $6 + \sqrt{2}$

24. Rasika has several unbiased 12-sided dice, each with faces labelled 1 to 12. When rolling all the dice at once, the probability of rolling a 12 exactly once is equal to the probability of rolling no 12s. How many dice does Rasika have?

Rasika 有几枚质地均匀的 12 面骰子, 每枚骰子上的各个面从 1 到 12 进行编号。同时投掷全部骰子, 编号 12 恰好出现一次的概率等于编号 12 完全没有出现的概率。请问 Rasika 一共有几枚骰子?

(A) 8

(B) 9

(C) 10

(D) 11

(E) 12

25. A polynomial $p(x)$ satisfies the relation $p(x+1) = x^2 - x + 2p(6)$ for every real x . What is the sum of the coefficients of p ?

对于每个实数 x , 多项式 $p(x)$ 满足 $p(x+1) = x^2 - x + 2p(6)$ 。请问 p 的系数之和是多少?

(A) -40

(B) -6

(C) 12

(D) 40

(E) None of the previous 以上都不正确

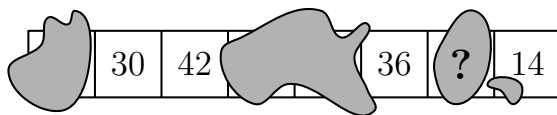
26. The values of x, y and z satisfy $2^x = 3$, $2^y = 7$ and $6^z = 7$. Which of the following gives the relationship between x, y and z ?

x, y, z 的值满足 $2^x = 3$, $2^y = 7$, $6^z = 7$ 。请问下列哪个选项表示 x, y, z 之间的关系?

- (A) $z = \frac{y}{1+x}$ (B) $z = \frac{x}{y} + 1$ (C) $z = \frac{y}{x} - 1$ (D) $z = \frac{x}{y-1}$ (E) $z = y - \frac{1}{x}$

27. A strip of paper consists of eight squares. Initially each square contains the number 0. In every move, we chose 4 consecutive squares and add one to each of the numbers in those squares. The figure shows the outcome after a number of moves but unfortunately some ink is covering some of the squares. What number is written on the square with the question mark?

一张纸条由 8 个方格组成。最初，每个方格中都包含数字 0。每一轮选择 4 个连续方格，并将所选方格中的每个数字加 1。图中所示为多轮操作后的结果，但其中一些方格被墨水遮盖。请问图中问号所在方格上的数字为多少？



- (A) 24 (B) 30 (C) 36 (D) 48
(E) None of the previous 以上都不正确

28. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(20-x) = f(22+x)$ for all real x . It is known that f has exactly two roots. What is the sum of these two roots?

对于所有实数 x ，函数 $f: \mathbb{R} \rightarrow \mathbb{R}$ 满足 $f(20-x) = f(22+x)$ 。已知 f 恰好有两个根。请问这两个根的和是多少？

- (A) -1 (B) 20 (C) 21 (D) 22
(E) None of the previous 以上都不正确

29. Twelve points are equally spaced on a circle. How many triangles containing a 45° angle can be formed by choosing three of these points?

一个圆上有 12 个等距分布的点。选取其中三个点，请问可以组成多少个含一个 45° 角的三角形？

- (A) 48 (B) 60 (C) 72 (D) 84 (E) 96

30. A special four-digit number $\overline{abcd}$ satisfies the equation $\overline{abcd} = a^a + b^b + c^c + d^d$. What is the value of a ?

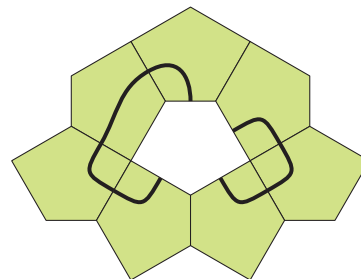
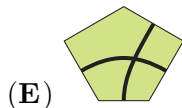
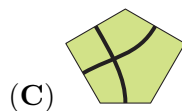
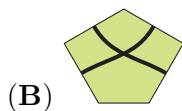
一个特殊的 4 位数 $\overline{abcd}$ 满足方程 $\overline{abcd} = a^a + b^b + c^c + d^d$ 。请问 a 的值是多少？

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

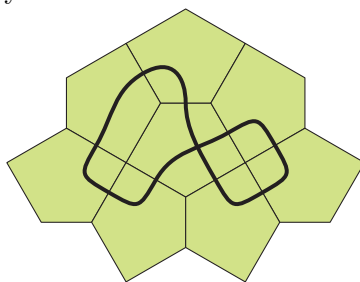
Student

3 points

1 (Slovenia). A pattern is made of equal pentagons. Which of the tiles below, when placed in the central hole, will form a self-intersecting loop?



SOLUTION: Note that all tiles are rotated by 180° . No other rotation can make the tile fit as the



pentagonal tile has exactly two right angles.

2 (United Kingdom). Which of these integers is two less than a multiple of ten, two more than a square, and two times a prime?

(A) 78

(B) 58

(C) 38

(D) 18

(E) 6

SOLUTION: All the given options but (E) are two less than a multiple of ten. However, both 38 and 18 are two more than a square (36 and 16) but only 38 is two times a prime (2×19). Hence the answer is 38.

3 (Germany). A young kangaroo cut a pizza into six equal slices. After eating one slice, he arranged the remaining slices with equal gaps between slices. What size is the angle of each gap?

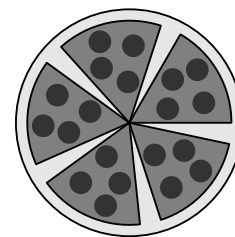
(A) 5°

(B) 8°

(C) 9°

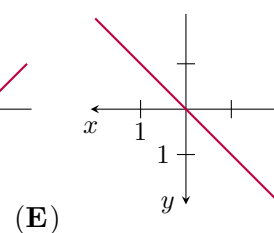
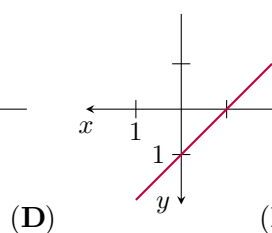
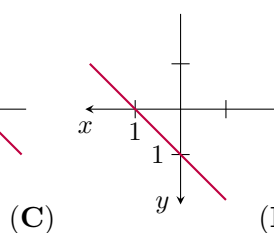
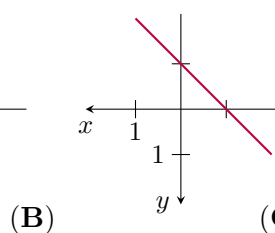
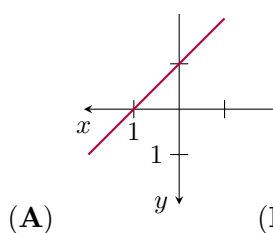
(D) 10°

(E) 12°



SOLUTION: The sum of the angles of all 5 gaps is $\frac{360^\circ}{6} = 60^\circ$. Hence, the angle of each gap is $\frac{60^\circ}{5} = 12^\circ$.

4 (Finland). Juuso has an unusual habit of drawing the xy -plane with the positive coordinate axes pointing left and down. What would the graph of the equation $y = x + 1$ look like in a coordinate system drawn by Juuso?



SOLUTION: The graph should pass the y -coordinate at 1 and y should increase as the x -coordinate increases, so the correct answer is D.

5 (Germany). Kaito has manipulated a die. The probabilities of rolling a 2, 3, 4 or 5 are still $\frac{1}{6}$ each, but the probability of rolling a 6 is twice the probability of rolling a 1. What is the probability of rolling a 6?

- (A) $\frac{1}{4}$ (B) $\frac{1}{6}$ (C) $\frac{7}{36}$ (D) $\frac{2}{9}$ (E) $\frac{5}{18}$

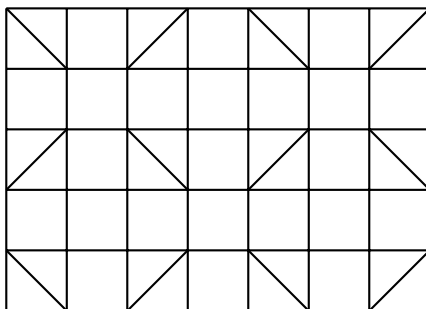
SOLUTION: We have $p(1)+p(6) = 1-4\cdot\frac{1}{6} = \frac{1}{3}$ and $p(6) = 2\cdot p(1)$. So we get $p(6) = \frac{1}{3}-p(1) = \frac{1}{3}-\frac{p(6)}{2}$. Therefore, $p(6) = \frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9}$.

6 (Russia). Which of the expressions below has the same value as $16^{15} + 16^{15} + 16^{15} + 16^{15}$?

- (A) 16^{19} (B) 4^{31} (C) 4^{60} (D) 16^{60} (E) 4^{122}

SOLUTION: Notice that $16 = 4^2$ and write the expression as follows: $16^{15} + 16^{15} + 16^{15} + 16^{15} = 4 \cdot (4^2)^{15} = 4 \cdot 4^{30} = 4^{31}$

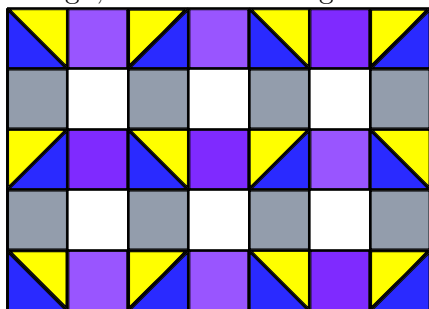
7 (Finland). Beaver wishes to color the squares and triangles of the following figure so that no two neighbouring figures, even those sharing a single vertex, are the same color.



What is the least number of colors needed?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

SOLUTION: There are points where five figures meet, so at least five colours are needed. Five is also enough, as seen in this figure.



8 (China). There are 6 glasses on a table with their open ends up. In any one move, we turn over exactly 4 of them. What is the least number of moves required to have all glasses upside down?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

SOLUTION: Whatever the first move, we will have 4 glasses down and 2 up. There is no 2nd move which will have all the glasses down. However we can move D(DDDU)U to D(UUUD)U which leaves 4 glasses pointing up which we can turn over in the third move. Hence the answer is 3.

9 (Greece). A student started with the number 1 and multiplied it by either 6 or 10. He then multiplied the result by either 6 or 10, and continued this procedure many times. Which of the following cannot be one of the numbers he obtained?

(A) $2^{100}3^{20}5^{80}$

(B) $2^{90}3^{20}5^{80}$

(C) $2^{90}3^{20}5^{70}$

(D) $2^{110}3^{80}5^{30}$

(E) $2^{50}5^{50}$

SOLUTION: If he used the factor 6 N times and the factor 10 M times, the number he would get is of the form $(2 \cdot 3)^N(2 \cdot 5)^M = 2^{N+M}3^N5^M$. Observe that the exponent of 2 is equal to the sum of the exponents of 3 and of 5. Of the numbers given the only one that fails this is B, as $90 \neq 20 + 80$.

10 (Greece). A black trail and a grey trail cross a park, as shown. Each trail divides the park into two regions of equal area. Which of the following must be true about the areas A , B and C ?

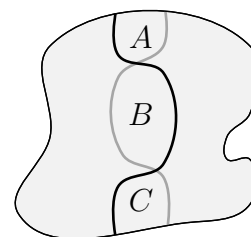
(A) $A = C$

(B) $B = A + C$

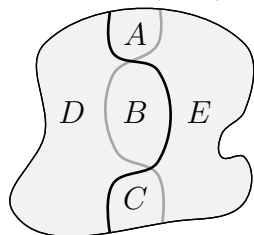
(C) $B = \frac{1}{2}(A + C)$

(D) $B = \frac{2}{3}(A + C)$

(E) $B = \frac{3}{5}(A + C)$



SOLUTION: To the left of the black trail, we have $D + B$ is half the park. To the left of the grey trail, we have $D + A + C$ is half the park. Equating these, we see that $B = A + C$.



4 points

11 (Finland). Exactly one of these statements about a certain positive integer n is true. Which statement is true?

(A) n is divisible by 3

(B) n is divisible by 6

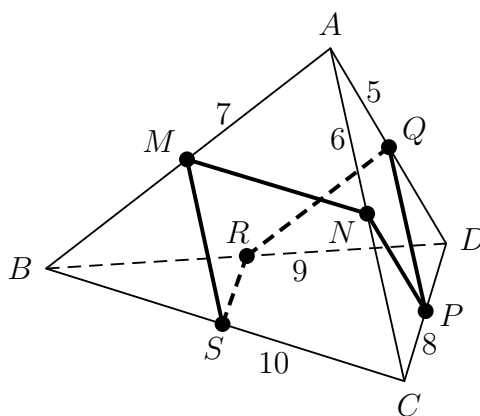
(C) n is odd

(D) $n = 2$

(E) n is prime

SOLUTION: D cannot be true since E would also be true. B cannot be true because A would also be true. Since n is not 2, if E were true, C would also be true, so E cannot be true. The true statement is either A or C. If A is true, then C is not, but then B would be true. So A cannot be true either, and the answer must be C. The number n must be some odd number that is not divisible by 3 nor is it prime, for example $n = 25$.

12 (Greece). A triangular pyramid $ABCD$ has sides of length 5, 6, 7, 8, 9 and 10. The points M , N , P , Q , R and S are the midpoints of the edges of the pyramid, as shown.



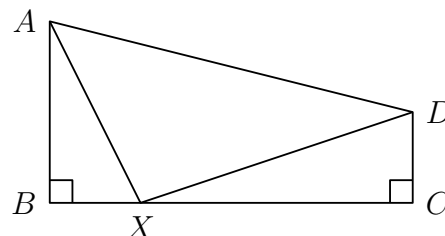
What is the perimeter of the closed hexagonal line $MNPQRSM$?

- (A) 19 (B) 20 (C) 21 (D) 22 (E) 23

SOLUTION: MN is half of BC . In fact each side of the closed hexagonal line is half a corresponding parallel edge. So the perimeter of the hexagon (starting from the segment MN) is $\frac{1}{2}(10 + 5 + 6 + 7 + 8 + 6) = 21$.

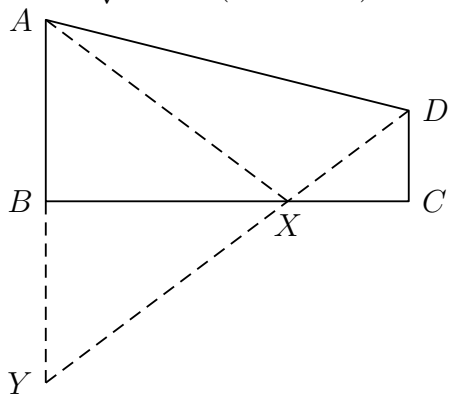
13 (China). A quadrilateral $ABCD$ has two right angles at B and C , where $AB = 4$, $BC = 8$ and $CD = 2$. Point X lies on BC . What is the minimum value of $AX + DX$?

- (A) $9\sqrt{2}$ (B) 12
(C) 13 (D) 10
(E) None of the previous



SOLUTION: If we extend the line AB to the point Y , where $AB = BY$, connect YD which crosses the side BC at point X , then connect AX . From the graph, we know that $AX = YX$, the minimum value of $AX + DX$ is YD .

$$YD = \sqrt{BC^2 + (BY + CD)^2} = \sqrt{8^2 + 6^2} = 10$$



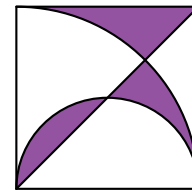
14 (Austria). John has a number of all black or all white unit cubes and wants to build a $3 \times 3 \times 3$ cube using 27 of them. He wants the surface to be exactly half black and half white. What is the smallest number of black cubes he can use?

- (A) 14 (B) 13 (C) 12 (D) 11
(E) None of the previous

SOLUTION: None of the numbers in the answers. In fact the smallest number is 10. There are 9 small squares showing on each of the 6 faces of the large cube, for a total of 54. In order for 27 of these to be black, he can put a black cube in each of the 8 corners ($3 \times 8 = 24$ small black squares) and one

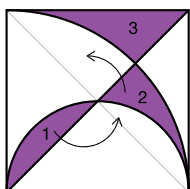
each in a central edge spot (2 black squares) and a central face spot (one black square).

15 (Russia). A diagonal, a semicircle and a quadrant are drawn in a square of side 6 cm. What is the area, in cm^2 , of the shaded part?



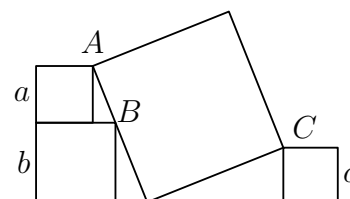
- (A) 9 (B) 3π (C) $6\pi - 9$
(D) $10\pi/3$ (E) 12

SOLUTION: Notice that part 1 coincides with the same part on the right. Combining of 1 and 2 coincides with the same part on the top:



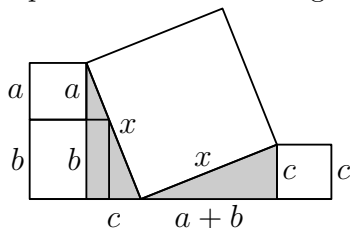
Thus all painted part coincides with the quarter of the square area. Then its area is $6^2 \div 4 = 36 \div 4 = 9$.

16 (Greece). The figure shows four squares. The smaller ones have side lengths a , b and c . The vertices A and C of two of the smaller squares coincide with two diagonally opposite vertices of the large square. The vertex B of the third small square is on the side of the large one. Which of the following expressions represents the side length of the largest square?



- (A) $\frac{1}{2}(a + b + c)$ (B) $\sqrt{a^2 + b^2 + c^2}$ (C) $\sqrt{(a + b)^2 + c^2}$
(D) $\sqrt{(b - a)^2 + c^2}$ (E) $\sqrt{a^2 + ab + b^2 + c^2}$

SOLUTION: It is easy to see that the two shaded right angled triangles are equal (hypotenuses equal and one acute angle equal). Then by Pythagoras on one of them we get $x^2 = (a + b)^2 + c^2$.



17 (United Kingdom). We have two positive numbers p and q , with $p < q$. Which of these expressions is the largest?

- (A) $\frac{p + 3q}{4}$ (B) $\frac{p + 2q}{3}$ (C) $\frac{p + q}{2}$ (D) $\frac{2p + q}{3}$ (E) $\frac{3p + q}{4}$

SOLUTION: First write all the options as expressions with a common denominator of 12 to give $\frac{3p + 9q}{12}$, $\frac{4p + 8q}{12}$, $\frac{6p + 6q}{12}$, $\frac{8p + 4q}{12}$ and $\frac{9p + 3q}{12}$. Since we are told that $p < q$ and the sum of the coefficients of p and q in each expression is the same, the largest expression is the one with the largest coefficient of q . Therefore the largest expression is $\frac{3p + 9q}{12}$ or $\frac{p + 3q}{4}$.

Alternatively, we can observe that each of the expressions is a weighted average of p and q . As $p < q$, the expression with the greatest proportion of q will be the largest, which is A.

18 (Russia). How many three-digit numbers are there that contain at least one of the digits 1, 2 or 3?

- (A) 27 (B) 147 (C) 441 (D) 557 (E) 606

SOLUTION: There are 900 three-digit numbers in total. Let's calculate how many of them do not contain 1, 2 or 3. The first digit of such a number could be 4, 5, 6, 7, 8 or 9, giving 6 options. For each of the second and third digits, there are 7 options. So there are $6 \cdot 7 \cdot 7 = 294$ numbers. Therefore there are $900 - 294 = 606$ numbers that contain at least one of digits 1, 2 or 3.

19 (Australia). I write down a 4-digit non-zero number $N = \overline{pqrs}$. When I place a decimal point between the q and the r , I find that the resulting number $\overline{pq.rs}$ is the average of the two-digit numbers $\overline{pq}$ and $\overline{rs}$.

What is the sum of the digits of N ?

- (A) 14 (B) 18 (C) 21 (D) 25 (E) 27

SOLUTION: We observe that pq and rs are whole numbers, so their average must end in .00 or .50. However, if $rs = 0$, the average of pq and rs cannot be more than pq , so rs is 50. This means that pq must be 49 and the average is 49.50. Hence the original number is 4950 with digital sum 18.

20 (Australia). Two candles of equal length start burning at the same time. One of the candles will burn down in 4 hours, the other in 5 hours, each at their own constant rate. How many hours will they have to burn before one candle is 3 times the length of the other?

- (A) $\frac{40}{11}$ (B) $\frac{45}{12}$ (C) $\frac{63}{20}$ (D) 3 (E) $\frac{47}{14}$

SOLUTION: Let the initial length of each candle be l . Let candle A be the faster burning candle, completely burning in 4 hours, and let B be the slower burning candle. After t hours the lengths of the candles are

A: $l(1 - t/4)$ B: $l(1 - t/5)$

We need to find t such that length of B = $3 \times$ length of A

$$l(1 - t/5) = 3l(1 - t/4)$$

$$20 - 4t = 60 - 15t$$

$$11t = 40$$

$$t = 40/11$$

5 points

21 (Czech Republic). Andre has six cards with one number written on each side of each card. The pairs of numbers on the cards are (5, 12), (3, 11), (0, 16), (7, 8), (4, 14) and (9, 10).

The cards can be placed in any order in the blank spaces of the figure.

$$\boxed{} + \boxed{} + \boxed{} - \boxed{} - \boxed{} - \boxed{} = ?$$

What is the smallest result he can get?

- (A) -23 (B) -24 (C) -25 (D) -26 (E) -27

SOLUTION: Let the numbers a , A , $a < A$, be written on one card and the numbers b , B , $b < B$, be written on another, and let be $a + A < b + B$. Then we observe that $a - B < b - A$. So we select the three cards with the smallest totals on both sides of the cards and put their smaller numbers to the + signs. Then we put the other three cards with their bigger numbers to the - signs .

22 (Australia). Kangaroo solves the equation $ax^2 + bx + c = 0$, and Beaver solves the equation $bx^2 + ax + c = 0$, where a, b, c are pairwise distinct non-zero integers. It turns out that the equations

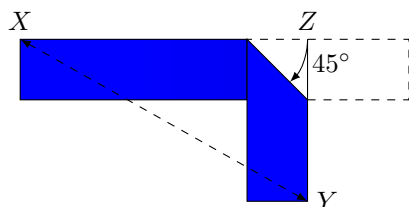
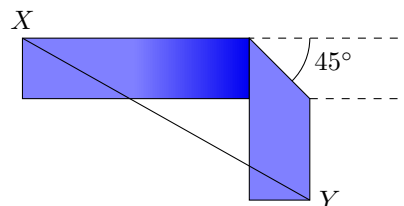
share a solution. Which of the following must be true?

- (A) The common solution must be 0.
 (B) The quadratic equation $ax^2 + bx + c = 0$ has exactly one real solution.
 (C) $a > 0$
 (D) $b < 0$
 (E) $a + b + c = 0$

SOLUTION: If the common solution is r , then we have $ar^2 + br + c = br^2 + ar + c$, which leads to $(a - b)r(r - 1) = 0$. Since $a \neq b$ and $c \neq 0$, it must be the case that $a - b \neq 0$ and $r \neq 0$. Hence, we deduce that the common solution is $r = 1$, which implies that $a + b + c = 0$.

23 (Australia). I have a strip of paper that is 12 cm long and 2 cm wide. I make a crease across it at 45° and then fold it, so that the two parts of the strip are aligned in a right angle, as shown. What is the smallest possible length, in cm, of XY ?

- (A) $6\sqrt{2}$ (B) $7\sqrt{2}$ (C) 10 (D) 8 (E) $6 + \sqrt{2}$



SOLUTION:

Consider the right-angled triangle XZY . The sides XZ and ZY add to 14. We want to minimise the hypotenuse XY of this triangle and this will occur when the triangle is isosceles. Hence the answer is $7\sqrt{2}$.

24 (Australia). Rasika has several unbiased 12-sided dice, each with faces labelled 1 to 12. When rolling all the dice at once, the probability of rolling a 12 exactly once is equal to the probability of rolling no 12s. How many dice does Rasika have?

- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

SOLUTION: Suppose she is using n dice. Rolling exactly one 12 amounts to choosing one of the n dice to show 12 and all $n - 1$ others to not show 12, which happens with probability $n \times \frac{1}{12} \times \left(\frac{11}{12}\right)^{n-1}$. On the other hand, rolling no 12s has probability $\left(\frac{11}{12}\right)^n$.

Setting these probabilities equal and cancelling a common factor of $\left(\frac{11}{12}\right)^{n-1}$, we see that $\frac{n}{12} = \frac{11}{12}$, so $n = 11$.

25 (Greece). A polynomial $p(x)$ satisfies the relation $p(x + 1) = x^2 - x + 2p(6)$ for every real x . What is the sum of the coefficients of p ?

- (A) -40 (B) -6 (C) 12 (D) 40
 (E) None of the previous

SOLUTION: Putting $x = 5$ we get $p(6) = 25 - 5 + 2p(6)$ so that $p(6) = -20$. Now the given equation becomes $p(x + 1) = x^2 - x - 2 \times 20 = x^2 - x - 40$ (*). A quick way to find the sum of the coefficients of p is to observe that it is $p(1)$ (this is true for all polynomials). So putting $x = 0$ we get $p(1) = -40$.

26 (Australia). The values of x, y and z satisfy $2^x = 3$, $2^y = 7$ and $6^z = 7$. Which of the following gives the relationship between x, y and z ?

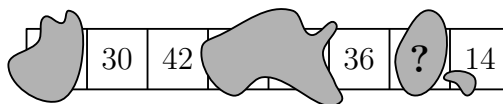
- (A) $z = \frac{y}{1+x}$ (B) $z = \frac{x}{y} + 1$ (C) $z = \frac{y}{x} - 1$ (D) $z = \frac{x}{y-1}$ (E) $z = y - \frac{1}{x}$

SOLUTION: Using various index laws, we have

$$7 = 6^z = (2 \times 3)^z = 2^z \times 3^z = 2^z \times (2^x)^z = 2^z \times 2^{xz} = 2^{z+xz} = 2^{z(1+x)}.$$

But also $2^y = 7$, so $y = z(1+x)$ and therefore $z = \frac{y}{1+x}$

27 (Greece). A strip of paper consists of eight squares. Initially each square contains the number 0. In every move, we chose 4 consecutive squares and add one to each of the numbers in those squares. The figure on the right shows the outcome after a number of moves but unfortunately some ink is covering some of the squares. What number is written on the square with the question mark?



- (A) 24 (B) 30 (C) 36 (D) 48
(E) None of the previous

SOLUTION: After a operations with the first four squares, b with the second run of four squares, etc, we will end up in a situation as in the figure.

A	B	C	D	E	F	G	H
a	$a+b$	$a+b+c$	$a+b+c+d$	$b+c+d+e$	$c+d+e$	$d+e$	e

From observing squares B and C , we see that $c = 12$. Now look at squares F and G . The difference between them is c , so $G = 36 - 12 = 24$.

If we want an example to show that the situation is attainable, we may take $a = 17, b = 13, c = 12, d = 10$ and $e = 14$.

28 (Greece). A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(20-x) = f(22+x)$ for all real x . It is known that f has exactly two roots. What is the sum of these two roots?

- (A) -1 (B) 20 (C) 21 (D) 22
(E) None of the previous

SOLUTION: If a is one of the roots we can choose x so that $20-x = a$, that is, $x = 20-a$. This gives $f(a) = f(42-a)$. So $42-a$ is also a root. Hence, the sum of the roots is $a + (42-a) = 42$.

Note that $42-a$ is definitely a different root because suppose $a = 42-a$, so $a = 21$. If b is the second root mentioned in the problem, so $b \neq 21$, then by the above $42-b$ is also a root. But as $b \neq 42-b$ (because $b \neq 21$), the equation would have three distinct roots, $a, b, 42-b$ contrary to assumption. Hence a and $42-a$ are distinct.

Another way to solve the problem is to use symmetry: The given equation can be written as $f(21-t) = f(21+t)$. So the line $x = 21$ is an axis of symmetry of the graph of f . The result follows from this.

29 (Australia). Twelve points are equally spaced on a circle. How many triangles containing a 45° angle can be formed by choosing three of these points?

- (A) 48 (B) 60 (C) 72 (D) 84 (E) 96

SOLUTION: An angle of 45° is created by a line subtended by two points spaced three apart. For any given base arc three points apart, there are 8 positions for the third vertex of the triangle. So we have $12 \times 8 = 96$ possible triangles. However, any isosceles triangles will be counted twice and there are 12 of these, so the total of different triangles is $96 - 12 = 84$

30 (Switzerland). A special four-digit number $\overline{abcd}$ satisfies the equation $\overline{abcd} = a^a + b^b + c^c + d^d$. What is the value of a ?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

SOLUTION: Estimation shows that $6^6 > 10000$, $4^4 = 256$ so we cannot have a 6 or more as one of the digits, but without the 5 the right hand side won't be larger than 1000, so among a, b, c, d there should be exactly one 5, that makes $a = 3$ since $5^5 = 3125$. The special number is 3435.